Total No. of Questions : 8]

SEAT No. :

PD4635

[Total No. of Pages : 2

[6404]-141

B.E. (Electronics & Telecommunication)

EMBEDDED SYSTEM DESIGN

(2019 Pattern) (Semester - VIII) (Elective - V) (404191(D))

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right side indicate full marks.*
4) *Use of Calculator is allowed.*
5) *Assume Suitable data if necessary.*

Q1) a) Describe the classification of API? **[9]**

b) Summarize the use of HAL library for SPI, 12C and CAN module. **[8]**

OR

Q2) a) Elaborate USB Modules in the STM 32F4 Microcontroller. **[8]**

b) Describe Hardware Abstraction Layer (HAL) drivers. **[9]**

Q3) a) Describe cross development compilation in detail. **[9]**

b) What is FreeRTOS? How FreeRTOS is configured using STM32CubeMX? **[9]**

OR

Q4) a) Identify in detail the role of GNU debugger. **[9]**

b) Describe ChibiOS and Contiki OS. **[9]**

P.T.O.

Q5) a) Elaborate how to Install Touch GFX for Graphical User Interface (GUI). **[9]**

b) Design an embedded system for Image transfer between PC and STM32F4. **[8]**

OR

Q6) a) Illustrate procedure of GUI Formation with TouchGFX for any two applications. **[8]**

b) Summarize how to Interface of SPI based graphical LCD with STM32F4. **[9]**

Q7) a) Describe Loading and interfacing methods in Embedded Android. **[9]**

b) Describe Linux Kernel. **[9]**

OR

Q8) a) Classify different android platforms used in embedded system in detail. **[9]**

b) Articulate requirments of android. **[9]**

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